HW1

Submit 8 problems total from the union of *LADW* **Ch 1.1-4** (copied below) and the supplementary problems.

Please label all *LADW* problems by "Chapter.Section.Exercise," e.g. 1.3.4. Label the supplementary problems as they're labeled on the handout.

Please put a "G" in front of the four problems you would like graded for correctness. Additionally, put an "F" in front of any of the problems graded for completion on which you would like written feedback from the grader.

Suggested *LADW* problems: 1.1.2, 1.1.7 or 1.1.8, 1.2.4, 1.2.6, 1.3.3 or 1.3.4, and 1.3.5

Section 1.1

Exercises.

1.1. Let $\mathbf{x} = (1,2,3)^T$, $\mathbf{y} = (y_1,y_2,y_3)^T$, $\mathbf{z} = (4,2,1)^T$. Compute $2\mathbf{x}$, $3\mathbf{y}$, $\mathbf{x}+2\mathbf{y}-3\mathbf{z}$.

1.2. Which of the following sets (with natural addition and multiplication by a scalar) are vector spaces. Justify your answer.

a) The set of all continuous functions on the interval $[0,1]$;
b) The set of all non-negative functions on the interval $[0,1]$;
c) The set of all polynomials of degree *exactly* n;
d) The set of all symmetric $n\times n$ matrices, i.e. the set of matrices $A = \{a_{j,k}\}_{j,k=1}^n$ such that $A^T = A$.

1.3. True or false:

a) Every vector space contains a zero vector;
b) A vector space can have more than one zero vector;
c) An $m\times n$ matrix has m rows and n columns;
d) If f and g are polynomials of degree n, then $f+g$ is also a polynomial of degree n;
e) If f and g are polynomials of degree at most n, then $f+g$ is also a polynomial of degree at most n

1.4. Prove that a zero vector $\mathbf{0}$ of a vector space V is unique.

1.5. What matrix is the zero vector of the space $M_{2\times 3}$?

1.6. Prove that the additive inverse, defined in Axiom 4 of a vector space is unique.

1.7. Prove that $0\mathbf{v} = \mathbf{0}$ for any vector $\mathbf{v}\in V$.

1.8. Prove that for any vector $\mathbf{v}$ its additive inverse $-\mathbf{v}$ is given by $(-1)\mathbf{v}$.

Section 1.2

Exercises.

2.1. Find a basis in the space of 3×2 matrices $M_{3\times 2}$.

2.2. True or false:

a) Any set containing a zero vector is linearly dependent
b) A basis must contain $\mathbf{0}$;
c) subsets of linearly dependent sets are linearly dependent;
d) subsets of linearly independent sets are linearly independent;
e) If $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \ldots + \alpha_n\mathbf{v}_n = \mathbf{0}$ then all scalars α_k are zero;

2.3. Recall, that a matrix is called *symmetric* if $A^T = A$. Write down a basis in the space of *symmetric* 2×2 matrices (there are many possible answers). How many elements are in the basis?

2.4. Write down a basis for the space of

a) 3×3 symmetric matrices;
b) $n\times n$ symmetric matrices;
c) $n\times n$ *antisymmetric* ($A^T = -A$) matrices;

2.5. Let a system of vectors $\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_r$ be linearly independent but not generating. Show that it is possible to find a vector $\mathbf{v}_{r+1}$ such that the system $\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_r,\mathbf{v}_{r+1}$ is linearly independent. **Hint:** Take for $\mathbf{v}_{r+1}$ any vector that cannot be represented as a linear combination $\sum_{k=1}^r \alpha_k\mathbf{v}_k$ and show that the system $\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_r,\mathbf{v}_{r+1}$ is linearly independent.

2.6. Is it possible that vectors $\mathbf{v}_1,\mathbf{v}_2,\mathbf{v}_3$ are linearly dependent, but the vectors $\mathbf{w}_1 = \mathbf{v}_1+\mathbf{v}_2$, $\mathbf{w}_2 = \mathbf{v}_2+\mathbf{v}_3$ and $\mathbf{w}_3 = \mathbf{v}_3+\mathbf{v}_1$ are linearly *independent*?

Section 1.3

Exercises.

3.1. Multiply:

a) $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$;

b) $\begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 3 \end{pmatrix}$;

c) $\begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$;

d) $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$.

3.2. Let a linear transformation in $\mathbb{R}^2$ be the reflection in the line $x_1 = x_2$. Find its matrix.

3.3. For each linear transformation below find it matrix

a) $T:\mathbb{R}^2\to\mathbb{R}^3$ defined by $T(x,y)^T = (x+2y, 2x-5y, 7y)^T$;
b) $T:\mathbb{R}^4\to\mathbb{R}^3$ defined by $T(x_1,x_2,x_3,x_4)^T = (x_1+x_2+x_3+x_4, x_2-x_4, x_1+3x_2+6x_4)^T$;
c) $T:\mathbb{P}_n\to\mathbb{P}_n$, $Tf(t) = f'(t)$ (find the matrix with respect to the standard basis $1,t,t^2,\ldots,t^n$);
d) $T:\mathbb{P}_n\to\mathbb{P}_n$, $Tf(t) = 2f(t)+3f'(t)-4f''(t)$ (again with respect to the standard basis $1,t,t^2,\ldots,t^n$).

3.4. Find 3×3 matrices representing the transformations of $\mathbb{R}^3$ which:

a) project every vector onto x-y plane;
b) reflect every vector through x-y plane;
c) rotate the x-y plane through $30°$, leaving z-axis alone.

3.5. Let A be a linear transformation. If $\mathbf{z}$ is the center of the straight interval $[\mathbf{x},\mathbf{y}]$, show that $A\mathbf{z}$ is the center of the interval $[A\mathbf{x}, A\mathbf{y}]$. **Hint:** What does it mean that $\mathbf{z}$ is the center of the interval $[\mathbf{x},\mathbf{y}]$?

3.6. The set $\mathbb{C}$ of complex numbers can be canonically identified with the space $\mathbb{R}^2$ by treating each $z = x+iy\in\mathbb{C}$ as a column $(x,y)^T\in\mathbb{R}^2$.

a) Treating $\mathbb{C}$ as a complex vector space, show that the multiplication by $\alpha = a+ib\in\mathbb{C}$ is a linear transformation in $\mathbb{C}$. What is its matrix?
b) Treating $\mathbb{C}$ as the real vector space $\mathbb{R}^2$ show that the multiplication by $\alpha = a+ib$ defines a linear transformation there. What is its matrix?
c) Define $T(x+iy) = 2x-y+i(x-3y)$. Show that this transformation is not a linear transformation in the complex vectors space $\mathbb{C}$, but if we treat $\mathbb{C}$ as the real vector space $\mathbb{R}^2$ then it is a linear transformation there (i.e. that T is a *real linear* but not a *complex linear* transformation).
Find the matrix of the real linear transformation T.

3.7. Show that any linear transformation in $\mathbb{C}$ (treated as a complex vector space) is a multiplication by $\alpha\in\mathbb{C}$.

Section 1.4

No exercises in this section.

Supplementary problems for HW1

I. How many vectors are there in the vector space $(\mathbb{Z}/5\mathbb{Z})^2$? How many (unordered) bases does this vector space have? Now answer the same questions for $(\mathbb{Z}/p\mathbb{Z})^n$.

II. For each of the following statements, decide if it is always, sometimes, or never true. If it is always or never true, give a proof. If it is sometimes true, give an example where it's true and where it's false. (In the future, I'll abbreviate all that by just: **ASN**)

Let V be a finite-dimensional vector space and let B_1 and B_2 be distinct bases for V. (That is, B_1 and B_2 are not equal as sets.)

1. $B_1 \cap B_2$ is a basis for V.
2. $B_1 \cup B_2$ is a basis for V.
3. the span of $B_1 \cap B_2$ equals V.
4. the span of $B_1 \cup B_2$ equals V.

III. Let V be a vector space and let J and K each be a set of two vectors in V, in such a way that J and K have no elements in common. **ASN**:

1. $\operatorname{span}(J) = \operatorname{span}(K)$
2. $\operatorname{span} J$ and $\operatorname{span} K$ are disjoint.
3. If there exists a vector x where $\operatorname{span}(J \cup \{x\}) = \operatorname{span}(K \cup \{x\})$, then $\operatorname{span}(J) = \operatorname{span}(K)$.

IV. Prove or provide a counterexample: For any finite-dimensional vector space V and any non-zero vector $v \in V$, there exists a basis for V that contains v.

V. Find two matrices $A, B \in M^{\mathbb{R}}_{2\times 2}$ so that they are not scalar multiples of each other and so that each commutes with the matrix $S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Then consider: do all linear combinations of A and B also commute with S? Finally, **ASN**: if $A, B, C \in M^{\mathbb{F}}_{n\times n}$ and both A and B commute with C, then $(kA + \ell B)$ commutes with C for all $k, \ell \in \mathbb{F}$.